

# Zheyuan Dai

Madison, WI

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## EDUCATION

### University of Wisconsin–Madison

Madison, WI

*M.S. in Mathematics, expected May 2027*

*Sep. 2026 – May 2027*

– Visiting International Student in Mathematics, Sep. 2025 – May 2026.

### Shanghai Jiao Tong University

Shanghai, China

*B.S. in Mathematics and Applied Mathematics; GPA: 3.40/4.00 Ranking: 21/47*

*Sep. 2022 – Jun. 2026*

## PUBLICATIONS

**Zheyuan Dai**, Xuliang Zhu, Yizhang He, Kai Wang, Fan Zhang, and Xuemin Lin. “Robust Coreness Estimation via H-index under Edge-LDP.” Manuscript under minor revision for *ACM SIGMOD 2027*.

**Zheyuan Dai**, Xuliang Zhu, Ruijia Wu, Yizhang He, Gengda Zhao, and Kai Wang. “Private Clicks, Public Actions: Logistic Bandits under Local Differential Privacy.” Submitted to *AAAI 2027*.

## RESEARCH EXPERIENCE

### Robust Coreness Estimation under Edge-Level Local Differential Privacy

*2025 – Present*

– Developed and analyzed iterative H-index estimators for private coreness estimation, with approximate  $k$ -core guarantees and evaluation on large real-world graphs.

### Logistic Bandits under Reward-Only Local Differential Privacy

*2026 – Present*

– Developed frequentist and Bayesian bandit algorithms with regret guarantees for private online recommendation and evaluated them on MovieLens and Last.fm.

### RepCut for Chip Design Optimization

*2026 – Present*

– Developed graph and hypergraph formulations for replication-aware partitioning in large-scale chip design.

### Monochromatic $K_4$ Minimization

*Oct. 2024*

– Implemented a conditional-expectation algorithm to reduce monochromatic  $K_4$  subgraphs in two-colored graphs.

## INDUSTRY EXPERIENCE

### Huawei Technologies Co., Ltd.

China

*Wireless Mobile Automation Engine (MAE) Intern*

*2025*

– Designed backup-power allocation algorithms for cellular base stations during outages and formulated resource-allocation strategies to extend network coverage under limited energy capacity.

## LEADERSHIP AND ACTIVITIES

### Shanghai Jiao Tong University Mountaineering Association

Shanghai, China

*President*

*Jun. 2023 – Jun. 2025*

– Led the university mountaineering team and organized technical training, expedition planning, and outdoor activities.

## HONORS

Outstanding Undergraduate Thesis Award, School of Mathematical Sciences, Shanghai Jiao Tong University, 2026.

Second Prize and Best Creativity Award, VEX Mechanical Competition, Shanghai Jiao Tong University, Nov. 2023.

## SKILLS

**Programming:** Python, C++, MATLAB, R, LaTeX    **Libraries:** PyTorch, pandas, NumPy, NetworkX

**Research Areas:** graph algorithms, differential privacy, statistical learning, stochastic processes, numerical methods

**Languages:** Chinese (native), English (TOEFL 102; CET-6)